KETAMINE

Preserves airway reflexes

Emergence reactions

CHEMICAL NAME (RS)-2-(2-chlorophenyl)-2-(methylamino)cyclohexan-1-one (as hydrochloride)
CLASS Dissociative anaesthetic (NMDA receptor antagonist)
TRADE NAMES Ketalar · Ketanest · Ketamine HCl injection

THERAPEUTIC USES / INDICATIONS

- Induction of anaesthesia, especially in shock, hypotension or asthma.
- Procedural sedation and analgesia (children, burns dressings, fracture reduction).
- Analgesia in trauma and opioid-sparing post-operative pain (low dose).
- Refractory status asthmaticus, treatment-resistant depression (esketamine).

ADMINISTRATION

- **IV:** slowly over 60 seconds; **IM** when IV access not available.
- Give in a quiet environment to reduce emergence reactions.
- Benzodiazepine premedication may reduce hallucinations.
- Controlled drug in many countries.

DOSAGES

- **Induction IV:** 1–2 mg/kg; **IM:** 4–10 mg/kg.
- **Procedural sedation IV:** 1 mg/kg, repeat 0.5 mg/kg.
- **Low-dose analgesia (off-label):** sub-anaesthetic doses per local protocol.
- **Children IM:** 4–5 mg/kg.

PHARMACOKINETICS

- **Onset:** IV 30 s; IM 3–4 min. **Duration:** IV 10–15 min; IM 15–30 min.
- Metabolised in liver to norketamine (active).
- Excreted in urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- Blocks **NMDA receptors** → dissociative anaesthesia (patient appears awake, eyes open, nystagmus, but unresponsive).
- Strong analgesia; sympathetic stimulation → ↑ HR and BP.
- Bronchodilation; airway reflexes and breathing largely preserved.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hypertension, tachycardia.
- Increased salivation; nystagmus.
- Nausea, vomiting on recovery.

ADVERSE EFFECTS (SERIOUS)

- Emergence reactions: vivid dreams, hallucinations, agitation (more in adults).
- Laryngospasm (rare); raised intraocular pressure.
- Raised ICP (traditional concern; less significant with ventilation).
- Misuse; bladder damage (chronic recreational use).

CONTRAINDICATIONS & CAUTIONS

- Conditions where BP rise is dangerous: severe hypertension, pre-eclampsia/eclampsia, recent MI, aortic dissection.
- Psychosis (schizophrenia).
- **Caution:** raised ICP, globe injury, thyrotoxicosis, porphyria.

KEY INTERACTIONS

- **Benzodiazepines:** reduce emergence reactions but prolong recovery.
- **Theophylline:** ↓ seizure threshold.
- **Thyroid hormones:** hypertension, tachycardia.

NURSING MANAGEMENT

- Monitor HR, BP, SpO_2, RR and airway; suction ready for secretions.
- Recover in a quiet, dim area with minimal stimulation.
- Observe for agitation or hallucinations; reassure; keep safe.
- Children: keep NBM per sedation guideline.
- Assess pain relief.

PATIENT EDUCATION

- You may have strange dreams or feel detached — this passes.
- Do not drive or drink alcohol for 24 h.
- Tell staff about any history of mental illness.

KEY POINTS

- NMDA antagonist; dissociative anaesthetic.
- Emergence reactions; quiet recovery.
- Maintains BP and airway reflexes — useful in shock and asthma.
- Avoid in pre-eclampsia, severe hypertension.

SOURCES CHECKED **[82]** Ketamine 10 mg/ml Injection (UK SmPC) **[83]** Ketamine Hydrochloride, IM, IV (US label)
[84] Ketamine Hydrochloride, IM, IV (US label, Medical Purchasing Solutions)

Full references p. 432–457